

End-Term Exam, May 09, 2025
Complex Analysis, Spring 2025

Max. Points: 50

Duration: 3 hours

Notes:

- Justify all relevant steps in detail.
- It is not necessary to use the given hint. You may ignore it if you prefer another method.

Notation: $\mathbb{C}$ denotes the set of all complex numbers.

1. (3 points) Show that the function $f(z) = e^x e^{-iy}$ is not holomorphic at any point of the complex plane.
2. (2+2 points) Locate and classify the isolated singularities of the following functions. (Do not ignore ∞ as a possible singularity.)

(a) $\frac{1}{z - z^3}$

(b) ze^{-z}

3. (3+3 points) Determine fractional linear maps which map:
 - (a) $i, -1, 1$ to $1, 0, \infty$
 - (b) $0, 1, \infty$ to $-1, 0, 1$
4. (4 points) Suppose the function $f(z)$ is entire and $f'(z)$ is bounded on $\mathbb{C}$. Determine the explicit form of the function $f(z)$.
5. (4 points) Consider the following Bessel J -function of order k :

$$J_k(z) = \sum_{n=0}^{\infty} \frac{(-1)^n}{n! (n+k)!} \left(\frac{z}{2}\right)^{2n+k}$$

Prove that it satisfies the following differential equation (justify every step):

$$z^2 J_k''(z) + z J_k'(z) + (z^2 - k^2) J_k(z) = 0, \quad \forall z \in \mathbb{C}.$$

6. (4 points) Find the Laurent series expansion of the function

$$f(z) = \frac{1}{(z-1)(z-2)} \quad \text{in } 0 < |z| < 1.$$

7. (4 points) Show that

$$\int_0^{2\pi} e^{e^{i\theta}} d\theta = 2\pi.$$

(Hint: You may want to use Cauchy integral formula.)

8. (6 points) Use residue theorem to show that

$$\int_0^{2\pi} \frac{d\theta}{2 - \sin \theta} = \frac{2\pi}{\sqrt{3}}.$$

9. (4 points) Let

$$P(z) = (z - a_1)(z - a_2) \cdots (z - a_n),$$

where $a_1, a_2, \dots, a_n$ are the roots of the polynomial $P(z)$. Let γ be a closed path such that none of the roots lie on γ . Show that

$$\frac{1}{2\pi i} \int_{\gamma} \frac{P'(z)}{P(z)} dz = W_{\gamma}(a_1) + W_{\gamma}(a_2) + \cdots + W_{\gamma}(a_n),$$

where $W_{\gamma}(a_i)$ denotes the winding number.

10. (4 points) Let

$$P_n(z) = \sum_{k=0}^n \frac{z^k}{k!}.$$

Given R , prove that $P_n(z)$ has no zeros in the disc of radius R for all n sufficiently large.

(Hint: Compare it with a function that never vanishes.)

11. (3 points) Let $f(z)$ be a meromorphic function on $\mathbb{C}$ but not entire. Let $g(z) = e^{f(z)}$. Show that $g(z)$ is not meromorphic on $\mathbb{C}$.
(Assume that $f(z)$ has only one singularity.)

12. (4 points) Prove that

$$f(z) = -\frac{1}{2} \left(z + \frac{1}{z} \right)$$

is a conformal map from the half-disk $\{z = x + iy : |z| < 1, y > 0\}$ to the upper half-plane.

(Hint: Use the representation $z = re^{i\theta}$ and analyze the image.)